

CHUQUR O'RGANISH USULLARI · Amaliy mashg'ulot (Practice)

5-mavzu. Attention mexanizmi va Transformer

2026–2027 o'quv yili · 30 ta topshiriq

Muhitni sozlash. Barcha kodlar Python 3.11/3.12 (CPU) va quyidagi qat'iy versiyalar bilan sinovdan o'tgan (`requirements.txt`); ma'lumotlar internetsiz: sintetik yoki topshiriq matnida berilgan.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# numpy==2.4.4 scipy==1.17.1 pandas==3.0.2 scikit-learn==1.8.0 scikit-image==0.26.0
# matplotlib==3.10.9 pillow==12.2.0 opencv-python-headless==4.13.0.92 torch==2.14.0 torchvision==0.29.0
```

Hujjatlar: <https://docs.pytorch.org/docs/stable/generated/torch.nn.MultiheadAttention.html>,
https://docs.pytorch.org/docs/stable/generated/torch.nn.functional.scaled_dot_product_attention.html,
<https://docs.pytorch.org/docs/stable/generated/torch.nn.TransformerEncoderLayer.html>.

Mavzu. Scaled dot-product attention, niqoblar (causal, padding), self-/multi-head attention, positional encoding, LayerNorm, Transformer bloki, murakkablik va KV-cache, dekodlash, BLEU va perplexity.

Maqsadlar. Attention'ni qo'lda va NumPy'da hisoblash; $1/\sqrt{d_k}$ va softmax to'yinishini isbotlash va o'lchash; attention orqali gradientni chiqarish; multi-head attention va PE ni noldan yozib PyTorch bilan solishtirish; niqob testini o'tkazish; kichik Transformer encoder va decoder (GPT) ni o'qitish.

Baholash (100 ball). 1–16: har biri 2 ball (32); 17–28: har biri 4 ball (48); 29, 30: har biri 10 ball (20). Kod topshiriqlarida: to'g'ri ishlashi 60%, talab qilingan chiqish formati 20%, natija talqini 20%. Umumiy talab: har bir kod topshirig'i alohida funksiya; `torch.manual_seed(42)`, `np.random.default_rng(42)`, CPU; `nn.TransformerEncoder(..., enable_nested_tensor=False)`; dastur `python -W error::DeprecationWarning` bilan ogohlantirishsiz ishlashi shart.

A. Qo'lda hisoblash, isbot va tushunchalar (1–16)

- Scaled dot-product attention.** $Q = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $V = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $d_k = 2$. $S = QK^T/\sqrt{d_k}$, $A = \text{softmax}(S)$ va AV ni toping.
- Causal mask.** $S = \begin{pmatrix} 0.5 & 1 & 2 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{pmatrix}$ (masshtablangan), $V = \begin{pmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 1 \end{pmatrix}$. Causal A va AV ni toping. 2-satrdagi niqobni softmax'dan *keyin* qo'llash (keraksiz og'irlikni nollash) qanday xato beradi?
- Self-attention.** $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $W^Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $W^K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $W^V = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, niqobsiz. Q, K, V, A va chiqishni toping. QK^T simmetrikmi?
- $1/\sqrt{d_k}$. q_i, k_i mustaqil, $\mathbb{E} = 0$, $\text{Var} = 1$ bo'lsa, $\text{Var}(\mathbf{q}^T \mathbf{k}) = d_k$ ekanini isbotlang. $d_k = 64$ va $d_k = 256$ da masshtablanmagan ballarning standart chetlanishi qancha?
- Softmax yakobiani.** $\partial a_i / \partial s_j = a_i(\delta_{ij} - a_j)$, ya'ni $J = \text{diag}(\mathbf{a}) - \mathbf{a}\mathbf{a}^T$ ekanini chiqaring; $J\mathbf{1} = \mathbf{0}$ ni ko'rsating. $\mathbf{s} = (1; 0; -1)$ uchun J va $\|J\|_F$ ni hisoblang; $\mathbf{a}$ one-hot bo'lsa $J = ?$
- Permutatsiya.** Niqobsiz self-attention uchun $\text{SA}(PX) = P\text{SA}(X)$ ni isbotlang. Nega bundan positional encoding zarurligi kelib chiqadi?
- Multi-head attention hajmi.** $d = 128$, $h = 8$: d_k , parametrlar soni ($4d^2 + 4d$) va $B = 4$, $T = 256$ da barcha attention matritsalaridagi elementlar soni. Parametrlar h ga bog'liqmi?
- Sinusoidal PE.** $d = 4$: PE_0, PE_2, PE_3 ni toping; $\|PE_{pos}\|$ va $PE_2^T PE_3 = PE_0^T PE_1$ ni tekshiring.
- Nisbiy siljish.** $PE_{pos+k} = R_k PE_{pos}$, R_k — blok-diagonal burish matritsasi ekanini isbotlang. $d = 4$ da ikkinchi juftlik ($\omega_1 = 0,01$) uchun R_1 ni PE_1 ga qo'llab PE_2 ni oling.
- LayerNorm.** $\mathbf{x} = (1; 3; 5; 7)$, $\gamma = 2$, $\beta = 0,5$, $\varepsilon \approx 0$: μ , σ^2 ($1/d$ bilan) va $\text{LN}(\mathbf{x})$.
- Blok va model parametrlari.** $d = 64$ bo'lgan bitta pre-norm blok ($h = 4$, FFN $4d$): $12d^2 + 13d$ ni tushuntiring va hisoblang. Model: $V = 50$, $d = 64$, $N = 2$, learned PE ($T_{\max} = 32$), yakuniy LN, bo'lishilmagan `Linear(d, V)`.
- Murakkablik.** a) T ni 256 dan 2048 ga oshirsak, attention matritsasi necha marta kattalashadi? b) $N = 24$, $d = 1024$, $T = 2048$, fp16 uchun KV-cache hajmi (bayt, MiB). c) $T = 100$, $d = 512$: $T^2 d$ va Td^2 ni solishtiring.

13. **Warmup.** $\eta(n) = d^{-1/2} \min(n^{-1/2}, n n_w^{-3/2})$, $d = 256$, $n_w = 4000$: $\eta(1000)$, $\eta(4000)$, $\eta(16000)$.
14. **Dekodlash.** $z = (3; 1; 0.5; 0)$: $\tau = 0.5; 1; 2$ da softmax; top- k ($k = 2$) va top- p ($p = 0.9$, $\tau = 1$) dan keyingi qayta normallangan taqsimotlar.
15. **BLEU.** Etalon: “men bugun kitob o'qidim”. a) Nomzod “kitob kitob o'qidim men”: clipped p_1 , p_2 , BP va BLEU-2. b) Nomzod “men kitob o'qidim”: p_1 , p_2 , BP, BLEU-2. Nega (b) qisqa bo'lsa ham jarimalanadi?
16. **Tushunchalar.** a) BERT, GPT, T5: arxitektura, niqob, o'qitish maqsadi. b) 256 tokenli ketma-ketlikda MLM: nechta token tanlanadi, nechitasi [MASK], tasodifiy, o'zgarishsiz? c) Nega padding niqobi *kalitlarga* qo'yiladi? d) Pre-norm va post-norm farqi; nega post-norm warmup talab qiladi?

B. Dasturlash (17–30)

Har bir funksiya natijani [T<nomer>] prefiksi bilan chop etadi; hammasini main() ishga tushiradi. NumPy tekshiruvlari float64 da; np.set_printoptions(precision=4, suppress=True).

17. **NumPy attention.** attention(Q, K, V, causal=False) -> (O, A) (satr maksimumini ayirib). 1-topshiriqni tekshiring. rng=np.random.default_rng(42); Q, K, V = (rng.normal(size=(5,8)) for _ in range(3)); F.scaled_dot_product_attention(..., is_causal=True) bilan maksimal farq, kelajakka berilgan eng katta vazn, satr yig'indilari.
18. **Masshtab va to'yinish.** Bitta rng (seed 42) bilan ketma-ket: $d_k \in \{4, 16, 64, 256, 1024\}$ uchun q , $k = \text{rng.normal(size=(2,20000,d_k))}$, $\text{Var}(q^T k)$ va $\text{Var}(q^T k / \sqrt{d_k})$; so'ng std $\in \{1, 4, 32\}$ uchun $\text{rng.normal}(0, \text{std}, (2000, 8))$ ballar: o'rtacha max a va o'rtacha $\|J\|_F$.
19. **Attention orqali backprop.** attention_backward(Q, K, V, G) -> (dQ, dK, dV) (ma'ruzadagi teorema). rng (seed 42): Q(4,3), K(5,3), V(5,2), G(4,2) — shu tartibda rng.normal; $L = \sum O \odot G$ uchun autograd bilan solishtiring. Ma'ruza mashqi: $Q = \begin{pmatrix} 1 & 0 \end{pmatrix}$, $K = I$, $V = \text{diag}(10, 20)$, $G = \begin{pmatrix} 1 & 0 \end{pmatrix}$ uchun dQ .
20. **Multi-head attention noldan.** class MyMHA(nn.Module): Wq, Wk, Wv, Wo = nn.Linear(d,d); view(B,T,h,dk).transpose(1,2). torch.manual_seed(42); ref=nn.MultiheadAttention(16,4,batch_first=True).double(), my=MyMHA(16,4).double(); og'irliklarni in_proj_weight/in_proj_bias bloklaridan va out_proj dan ko'chiring; x=torch.randn(2,7,16, dtype=torch.float64). Parametrlar soni, chiqishlar farqi va boshlar bo'yicha o'rtacha A farqi.
21. **PE xossalari.** sinusoidal_pe(T, d), rotation(k, d). $d = 4$ da PE[0..3] (8-topshiriq); $T = 100$, $d = 16$: normalar min/max, $\max_p |PE_{p+5} - R_5 PE_p|$, R_5 ortogonalligi, $PE_p^T PE_{p+5}$ min/max.
22. **Niqob testi.** Ma'ruzadagi pre-norm Block(d=32, h=4, causal) (attn_mask: True = yopiq). torch.manual_seed(42), nn.Sequential(nn.Embedding(29,32), Block(causal=...)). mask_test(model): g=Generator().manual_seed(0), a=randint(0,29,(1,10),generator=g), b=a.clone(), b[0,-1]=(a[0,-1]+1)%29; d_{pref} , d_{oxirgi} ni causal=True/False uchun va blok parametrlarini chiqaring.
23. **Padding niqobi.** torch.manual_seed(42), so'ng mha = nn.MultiheadAttention(8, 2, batch_first=True).eval(); x=torch.randn(1,6,8); x2=x.clone(); x2[0,4:]=torch.randn(2,8)*10; oxirgi 2 pozitsiya — padding (key_padding_mask). Niqob bilan va niqobsiz haqiqiy 4 token chiqishlari farqi; pad kalitlarga berilgan eng katta vazn.
24. **Blokli (online) softmax.** blockwise_attention(Q, K, V, block): K, V ni bloklab o'qib, (m, ℓ) va akkumulyatorni yangilang (FlashAttention g'oyasi). rng (seed 42): Q(6,8), K(10,8), V(10,3); block $\in \{1, 3, 4, 10\}$ da to'liq attention bilan farq.
25. **LayerNorm va parametrlar.** layer_norm(x, gamma, beta, eps=1e-5); 10-topshiriqni nn.LayerNorm bilan tekshiring. nn.TransformerEncoderLayer(64, 4, 256, batch_first=True, norm_first=True) parametrlari va 11-topshiriq modeli (TinyEncoderModel: token va pozitsiya Embedding, 2 qatlamli TransformerEncoder, LN, Linear) parametrlari hamda chiqish shakli (kirish zeros(2,32)).
26. **Temperature, top-k, top-p.** decode_probs(z, tau=1.0, top_k=None, top_p=None); 14-topshiriqni tekshiring. Top-p taqsimotidan torch.multinomial(p, 1000, replacement=True,

`generator=Generator().manual_seed(42))` bilan 1000 ta namuna oling va chastotalarni chiqaring.

27. BLEU. `bleu(cand, ref, N=4) -> (bleu, [p_1..p_N])` (bo'shliq bo'yicha tokenlash, clipping, BP, smoothing'siz). 15-topshiriqni ($N = 2$) tekshiring. Etalon "bugun ertalab men kutubxonada yangi kitob o'qidim" uchun nomzodlar "bugun ertalab men kutubxonada kitob o'qidim" va "men bugun kitob o'qidim" ning BLEU-4 ini hisoblang va izohlang.

28. Tiny Transformer encoder: ketma-ketlikni teskari aylantirish. Kirish — 0.9 raqamlardan $T = 8$ uzunlikdagi ketma-ketlik, nishon — `X.flip(1)`. Model: `Embedding(10,32) [+ learned PE Embedding(8,32)]`, `TransformerEncoder(TransformerEncoderLayer(32, 4, 64, dropout=0.0, batch_first=True, norm_first=True), 2, enable_nested_tensor=False)`, `Linear(32,10)`. Test: `Generator().manual_seed(123), randint(0,10,(1000,8))`. O'qitish: `torch.manual_seed(42)` dan keyin model; `g` (seed 42) bilan har qadamda yangi `randint(0,10,(64,8))`; Adam $3e-3$, 600 qadam, cross-entropy. PE bilan va PE siz: token va to'liq ketma-ketlik aniqligi. Natijani 6-topshiriq bilan izohlang.

29. Tiny GPT (decoder-only, char-level). Korpus (maqollar; 90% train, 10% val):

```
CORPUS = ("ilm olish igna bilan quduq qazish. sabr tagi sariq oltin. "
          "bilganing bilganing, bilmaganing bir tuki. yaxshi so'z jon ozig'i. "
          "mehnat qilsang rohat topasan. til qilichdan o'tkir. "
          "birlashgan o'zar, birlashmagan to'zar. ko'p o'qigan ko'p biladi. "
          "ilm olish igna bilan quduq qazish, sabr tagi sariq oltin. "
          "mehnat qilsang rohat topasan, ko'p o'qigan ko'p biladi. ")
```

TinyGPT: token `Embedding(V,48)` + pozitsiya `Embedding(64,48)`; 2 qatlamli `TransformerEncoder` ($d = 48$, $h = 4$, FFN 192, `dropout=0.1`, `norm_first=True`) causal niqob bilan (`nn.Transformer.generate_square_subsequent_mask(T)`, `is_causal=True`); `LN`; `Linear(48,V)`. `torch.manual_seed(42)`; AdamW ($lr=3e-3$, `weight_decay=0.01`); 500 qadam; 32 ta 48 belgili oyna (`randint(0, len(tr)-49, (32,))`, `generator=g`), `g` seed 42); clip 1,0. Har 100 qadamda train/val PPL (48 lik ketma-ket bo'laklar bo'yicha); yakunda niqob testi ($T = 20$) va "ilm " dan 40 belgi greedy generatsiya.

30. KV-cache. Bitta boshli causal attention: `causal_attention_full(X, Wq, Wk, Wv)` va tokenlarni birma-bir qayta ishlaydigan `causal_attention_kv_cache(...)`. `rng` (seed 42): `X(10,16)`, so'ng `Wq, Wk, Wv = rng.normal(size=(16,16))/4`. Farqni chiqaring; $T = 10$ va $T = 1024$ uchun ball hisoblashdagi ko'paytirishlar soni ($\sum_t td$ va keshsiz $\sum_t t^2d$) va GPT-2 small KV-cache hajmi ($N = 12$, $d = 768$, $T = 1024$, fp16).